

Vector Calculus Notes

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1 Introduction to Vector Calculus and Parametric Equations

Equation of a Straight Line Through (x_0, y_0, z_0) with Direction (l, m, n)

$$\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n} = t$$

Parametric form:

$$x = x_0 + lt, \quad y = y_0 + mt, \quad z = z_0 + nt$$

Vector form:

$$\vec{r}(t) = \vec{a} + \vec{b}t, \quad \text{where } \vec{a} = (x_0, y_0, z_0), \quad \vec{b} = (l, m, n)$$

Equation of a Line Passing Through Two Points (x_1, y_1, z_1) and (x_2, y_2, z_2)

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1} = t$$

$$\implies x = x_1 + (x_2 - x_1)t, \quad y = y_1 + (y_2 - y_1)t, \quad z = z_1 + (z_2 - z_1)t$$

Vector form:

$$(x, y, z) = \vec{r}(t) = (x_1, y_1, z_1) + [(x_2 - x_1), (y_2 - y_1), (z_2 - z_1)]t$$

$$\vec{r}(t) = \vec{a} + (\vec{b} - \vec{a})t$$

Tangent Vector and Tangent Line

Tangent vector:

$$\vec{r}'(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} [\vec{r}(t + \Delta t) - \vec{r}(t)]$$

Equation of tangent line to curve C at point P is given by:

$$\vec{a}(w) = \vec{r}_0 + w\vec{r}'_0$$

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2 Length of a Curve

Arc length differential ds and total length:

$$\frac{ds}{dt} = \int_a^b \left[\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 + \left(\frac{dz}{dt} \right)^2 \right]^{1/2} dt$$

$$s = \int_a^b \left| \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right) \right| dt = \int_a^b \left| \frac{d\vec{r}}{dt} \right| dt$$

Example 1: What curve is represented by $[3 + 6 \cos t, -2 + \sin t, 4]$?

Let $x = 3 + 6 \cos t$, $y = -2 + \sin t$, $z = 4$. Using $\cos^2 t + \sin^2 t = 1$:

$$\left(\frac{x-3}{6} \right)^2 + (y+2)^2 = 1, \quad z = 4 \quad (\text{An ellipse in the plane } z = 4)$$

Example 2: Represent the curve $4x^2 - 3y^2 = 12$, $z = 1$ in parametric form.

$$\frac{4x^2}{12} - \frac{3y^2}{12} = 1 \implies \frac{x^2}{3} - \frac{y^2}{4} = 1 \implies \frac{x^2}{(\sqrt{3})^2} - \frac{y^2}{2^2} = 1$$

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3 Tangent and Unit Tangent Vectors

Example: Given a curve $\vec{r}(t) = t\hat{i} + t^3\hat{j}$, find the tangent vector $\vec{r}'(t)$, unit tangent vector $\hat{u}(t)$ at $(1, 1, 0)$, and the equation of the tangent line at P .

At $(1, 1, 0) = (t, t^3, 0) \implies t = 1$.

$$\begin{aligned} \vec{r}(t) &= t\hat{i} + t^3\hat{j} \\ \vec{r}'(t) &= \hat{i} + 3t^2\hat{j} \\ \hat{u}(t) &= \frac{\vec{r}'(t)}{|\vec{r}'(t)|} = \frac{\hat{i} + 3t^2\hat{j}}{\sqrt{1 + 9t^4}} \end{aligned}$$

At $P(1, 1, 0)$ ($t = 1$):

$$\begin{aligned} \vec{r}'(1, 1, 0) &= \hat{i} + 3\hat{j} \\ \hat{u}(1, 1, 0) &= \frac{1}{\sqrt{10}}\hat{i} + \frac{3}{\sqrt{10}}\hat{j} \end{aligned}$$

Equation of the tangent line:

$$\vec{r}_{\text{tangent}} = \vec{r}_0 + w\vec{r}'_0 = (\hat{i} + \hat{j}) + w(\hat{i} + 3\hat{j}) = (1+w)\hat{i} + (1+3w)\hat{j}$$

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4 Gradient and Directional Derivative

Gradient

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

For $f(x, y, z) = xyz$:

$$\text{grad } f = yz\hat{i} + xz\hat{j} + xy\hat{k}$$

Directional Derivative

$$D_{\vec{a}}f = \frac{\partial f}{\partial s} = \lim_{s \rightarrow 0} \frac{f(Q) - f(P)}{s} \quad \text{in direction of } \vec{a}$$

Along $\vec{r}(s) = P_0 + s\vec{b}$, using the chain rule:

$$D_{\vec{b}}f = \frac{df}{ds} = \frac{\partial f}{\partial x} \frac{dx}{ds} + \frac{\partial f}{\partial y} \frac{dy}{ds} + \frac{\partial f}{\partial z} \frac{dz}{ds} = \left(\frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k} \right) \cdot \left(\frac{dx}{ds}\hat{i} + \frac{dy}{ds}\hat{j} + \frac{dz}{ds}\hat{k} \right)$$

$$D_{\vec{b}}f = \text{grad } f \cdot \frac{d\vec{r}}{ds} = \text{grad } f \cdot \vec{b} = \frac{\vec{b}}{|\vec{b}|} \cdot \text{grad } f$$

Example: Find the directional derivative of $f = 2x^2 + 3y^2 + z^2$ at $(2, 1, 3)$ in the direction $\vec{a} = \hat{i} - 2\hat{k}$.

$$\text{grad } f = 4x\hat{i} + 6y\hat{j} + 2z\hat{k}$$

$$\text{grad } f|_{(2,1,3)} = 8\hat{i} + 6\hat{j} + 6\hat{k}$$

$$\hat{a} = \frac{\hat{i} - 2\hat{k}}{\sqrt{1+4}} = \frac{\hat{i} - 2\hat{k}}{\sqrt{5}}$$

$$D_{\vec{a}}f = \frac{1}{\sqrt{5}}(\hat{i} - 2\hat{k}) \cdot (8\hat{i} + 6\hat{j} + 6\hat{k}) = \frac{8-12}{\sqrt{5}} = -\frac{4}{\sqrt{5}}$$

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5 Properties of Gradient and Normal Vectors

Maximum value of directional derivative:

$$D_{\vec{b}}f = |\vec{b}| |\text{grad } f| \cos \theta \implies \max D_{\vec{b}}f = |\text{grad } f| \quad (\text{when } \theta = 0)$$

Directional derivative has maximum value equal to $|\text{grad } f|$, so $\text{grad } f$ is the direction of maximum increase of f at point P .

Theorem: Let f be a differentiable function representing a surface $S : f(x, y, z) = c$ (constant). If $\text{grad } f \neq \vec{0}$ at P , then $\text{grad } f$ is normal to surface S at P .

Proof: Differentiating $f(x, y, z) = c$ with respect to parameter t along any curve on S :

$$\frac{df}{dt} = 0 \implies \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} = 0$$

$$\left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k} \right) \cdot \left(\frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k} \right) = 0 \implies \text{grad } f \cdot \frac{d\vec{r}}{dt} = 0$$

$\implies \text{grad } f \perp \frac{d\vec{r}}{dt}$, which means $\text{grad } f$ is normal to S .

6 Normal Vector, Laplacian, and Potential Functions

Example: Find the unit normal vector of $z^2 = 4(x^2 + y^2)$ at $(1, 0, 2)$.

Let $f(x, y, z) = 4(x^2 + y^2) - z^2$:

$$\begin{aligned} \text{grad } f &= 8x\hat{i} + 8y\hat{j} - 2z\hat{k} \\ \text{grad } f|_{(1,0,2)} &= 8\hat{i} + 0\hat{j} - 4\hat{k} \\ \text{Unit normal } \hat{n} &= \frac{\text{grad } f}{|\text{grad } f|} = \frac{8\hat{i} - 4\hat{k}}{\sqrt{64 + 16}} = \frac{4(2\hat{i} - \hat{k})}{\sqrt{80}} = \frac{2}{\sqrt{5}}\hat{i} - \frac{1}{\sqrt{5}}\hat{k} \end{aligned}$$

Laplacian Operator

$$\nabla^2 f = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) f$$

Product Rule for Gradient

$$\nabla(fg) = f\nabla g + g\nabla f$$

Example: Find $f(x, y)$ if $\vec{v} = \text{grad } f = (2x^3y^4 + x, 2x^4y^3 + y)$:

$$\frac{\partial f}{\partial x} = 2x^3y^4 + x \implies f(x, y) = \frac{1}{2}x^4y^4 + \frac{1}{2}x^2 + h(y)$$

Differentiating with respect to y :

$$\frac{\partial f}{\partial y} = 2x^4y^3 + h'(y) = 2x^4y^3 + y \implies h'(y) = y \implies h(y) = \frac{1}{2}y^2 + c$$

$$\implies f(x, y) = \frac{1}{2}x^4y^4 + \frac{1}{2}x^2 + \frac{1}{2}y^2 + c$$

7 Divergence of a Vector Field

Definition

For a vector field $\vec{v} = (v_1, v_2, v_3)$:

$$\text{div } \vec{v} = \nabla \cdot \vec{v} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (v_1, v_2, v_3) = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

Example 1: Find $\operatorname{div} \vec{v}$ for $\vec{v} = (2x^2z, -xy^2z, 3yz^2)$ at $(1, 1, 1)$.

$$\operatorname{div} \vec{v} = \frac{\partial}{\partial x}(2x^2z) + \frac{\partial}{\partial y}(-xy^2z) + \frac{\partial}{\partial z}(3yz^2) = 4xz - 2xyz + 6yz$$

At $(1, 1, 1)$:

$$\operatorname{div} \vec{v}|_{(1,1,1)} = 4(1)(1) - 2(1)(1)(1) + 6(1)(1) = 4 - 2 + 6 = 8$$

Example 2: Find $\operatorname{div} \vec{v}$ for $\vec{v} = (e^y, e^{-x}y, 2z \sinh x)$.

$$\begin{aligned} \operatorname{div} \vec{v} &= \frac{\partial}{\partial x}(e^y) + \frac{\partial}{\partial y}(e^{-x}y) + \frac{\partial}{\partial z}(2z \sinh x) \\ &= 0 + e^{-x} + 2 \sinh x \\ &= e^{-x} + 2 \left(\frac{e^x - e^{-x}}{2} \right) = e^{-x} + e^x - e^{-x} = e^x \end{aligned}$$

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8 Divergence Identities and Exercises

Identity 1: Prove $\operatorname{div}(k\vec{v}) = k \operatorname{div} \vec{v}$ for constant k .

$$\begin{aligned} k\vec{v} &= k(v_1, v_2, v_3) = (kv_1, kv_2, kv_3) \\ \operatorname{div}(k\vec{v}) &= \nabla \cdot (k\vec{v}) = \frac{\partial}{\partial x}(kv_1) + \frac{\partial}{\partial y}(kv_2) + \frac{\partial}{\partial z}(kv_3) \\ &= k \frac{\partial v_1}{\partial x} + k \frac{\partial v_2}{\partial y} + k \frac{\partial v_3}{\partial z} = k \left(\frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \right) = k \nabla \cdot \vec{v} = k \operatorname{div} \vec{v} \end{aligned}$$

Identity 2: Prove $\operatorname{div}(\operatorname{grad} f) = \nabla^2 f$.

$$\operatorname{div}(\operatorname{grad} f) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \nabla^2 f$$

Exercise 3: Find $\nabla^2 f$ if $f = \frac{xy}{z^2}$.

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{y}{z^2} \implies \frac{\partial^2 f}{\partial x^2} = 0 \\ \frac{\partial f}{\partial y} &= \frac{x}{z^2} \implies \frac{\partial^2 f}{\partial y^2} = 0 \\ \frac{\partial f}{\partial z} &= -\frac{2xy}{z^3} \implies \frac{\partial^2 f}{\partial z^2} = \frac{6xy}{z^4} \\ \nabla^2 f &= 0 + 0 + \frac{6xy}{z^4} = \frac{6xy}{z^4} \end{aligned}$$

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9 Curl of a Vector Field

For a 2D vector field $\vec{v} = [P(x, y), Q(x, y)]$:

- If $\frac{\partial P}{\partial y} < 0 \implies$ Curl is positive (+ve).
- If $\frac{\partial Q}{\partial x} > 0 \implies$ Curl is positive (+ve).

2D Curl scalar value:

$$\text{curl } \vec{v} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

3D Curl determinant definition:

$$\text{curl } \vec{v} = \nabla \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

10 Curl of a Vector Field

Definition of Curl

For a 3D vector field $\vec{V} = (V_1, V_2, V_3)$:

$$\text{Curl } \vec{V} = \nabla \times \vec{V} = \left(\frac{\partial V_3}{\partial y} - \frac{\partial V_2}{\partial z} \right) \hat{i} + \left(\frac{\partial V_1}{\partial z} - \frac{\partial V_3}{\partial x} \right) \hat{j} + \left(\frac{\partial V_2}{\partial x} - \frac{\partial V_1}{\partial y} \right) \hat{k}$$

Example 1: Given $\vec{V} = x^2yz\hat{i} + xy^2z\hat{j} + xyz^2\hat{k}$, evaluate Curl $\vec{V}$ at $(1, 2, 3)$.

$$\text{Curl } \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2yz & xy^2z & xyz^2 \end{vmatrix}$$

Evaluating the determinant:

$$= (xz^2 - xy^2) \hat{i} + (x^2y - yz^2) \hat{j} + (y^2z - x^2z) \hat{k}$$

At point $(1, 2, 3)$ ($x = 1, y = 2, z = 3$):

$$\text{Component } \hat{i} : 1(3^2) - 1(2^2) = 9 - 4 = 5$$

$$\text{Component } \hat{j} : 1^2(2) - 2(3^2) = 2 - 18 = -16$$

$$\text{Component } \hat{k} : 2^2(3) - 1^2(3) = 12 - 3 = 9$$

$$\text{Curl } \vec{V} \Big|_{(1,2,3)} = 5\hat{i} - 16\hat{j} + 9\hat{k}$$

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11 Vector Field Identities

1. Divergence of a Curl

Prove that $\text{div}(\text{curl } \vec{V}) = 0$:

$$\text{div}(\text{curl } \vec{V}) = \frac{\partial}{\partial x} \left(\frac{\partial V_3}{\partial y} - \frac{\partial V_2}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial V_1}{\partial z} - \frac{\partial V_3}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial V_2}{\partial x} - \frac{\partial V_1}{\partial y} \right)$$

Expanding using Clairaut's theorem ($\frac{\partial^2 V}{\partial x \partial y} = \frac{\partial^2 V}{\partial y \partial x}$):

$$= \frac{\partial^2 V_3}{\partial x \partial y} - \frac{\partial^2 V_2}{\partial x \partial z} + \frac{\partial^2 V_1}{\partial y \partial z} - \frac{\partial^2 V_3}{\partial y \partial x} + \frac{\partial^2 V_2}{\partial z \partial x} - \frac{\partial^2 V_1}{\partial z \partial y} = 0$$

2. Curl of a Gradient

Prove that $\text{Curl}(\text{grad } f) = \vec{0}$:

$$\text{grad } f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

$$\text{Curl}(\text{grad } f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix}$$

$$= \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \right) \hat{i} + \left(\frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} \right) \hat{j} + \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right) \hat{k} = \vec{0}$$

12 2D Curl Analysis and Line Integrals

2D Curl Evaluation Example

Given a 2D vector field $\vec{V} = (y^3 - 9y, x^3 - 9x)$:

$$\text{Curl } \vec{V} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = (3x^2 - 9) - (3y^2 - 9)$$

Case 1: At $x = 3, y = 0$:

$$(27 - 9) - (0 - 9) = 18 + 9 = 27 > 0 \quad (\text{Positive})$$

Case 2: At $x = 0, y = 3$:

$$(0 - 9) - (27 - 9) = -9 - 18 = -27 < 0 \quad (\text{Negative})$$

Case 3: At $x = 0, y = 0$:

$$(0 - 9) - (0 - 9) = -9 + 9 = 0$$

Line Integrals Along Different Paths

Evaluate $I = \int (x^2y \, dx + (x - 2y) \, dy)$ from $(0, 0)$ to $(1, 1)$:

Path 1: Along parabola $y = x^2$ Parametrization: $x = t, y = t^2 \implies dx = dt, dy = 2t \, dt$ for $t \in [0, 1]$.

$$\begin{aligned} I &= \int_0^1 \left[t^2(t^2) + (t - 2t^2)(2t) \right] dt = \int_0^1 \left[t^4 + 2t^2 - 4t^3 \right] dt \\ &= \left[\frac{t^5}{5} + \frac{2t^3}{3} - t^4 \right]_0^1 = \frac{1}{5} + \frac{2}{3} - 1 = -\frac{2}{15} \end{aligned}$$

Path 2: Along straight line $y = x$ Parametrization: $x = t, y = t \implies dx = dt, dy = dt$ for $t \in [0, 1]$.

$$\begin{aligned} I &= \int_0^1 \left[t^2(t) + (t - 2t) \right] dt = \int_0^1 (t^3 - t) dt \\ &= \left[\frac{t^4}{4} - \frac{t^2}{2} \right]_0^1 = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4} \end{aligned}$$

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13 Scalar and Vector Line Integrals

Scalar Line Integral Example

Evaluate $\int_C f(r) \, ds$ where $f = x^2 + y^2$ along $y = 3x$ from $(0, 0)$ to $(2, 6)$.

Parametrization: $x = t, y = 3t \implies \vec{r}(t) = t\hat{i} + 3t\hat{j}$.

$$\frac{d\vec{r}}{dt} = \hat{i} + 3\hat{j} \implies \left| \frac{d\vec{r}}{dt} \right| = \sqrt{1^2 + 3^2} = \sqrt{10}$$

Limits: $t \in [0, 2]$. Substituting $f(r(t)) = t^2 + (3t)^2 = 10t^2$:

$$\int_C f(r) \, ds = \int_0^2 10t^2 \sqrt{10} \, dt = 10\sqrt{10} \left[\frac{t^3}{3} \right]_0^2 = \frac{80\sqrt{10}}{3}$$

Vector Line Integral and Path Independence

Calculate $\int_C \vec{F}(r) \cdot d\vec{r}$ for $\vec{F} = [y^2, -x^2]$ along the line $(0, 0)$ to $(1, 4)$:

Parametrization: $x = t, y = 4t \implies \vec{r}(t) = t\hat{i} + 4t\hat{j}, \frac{d\vec{r}}{dt} = (1, 4)$. Limits: $t \in [0, 1]$.

$$\vec{F}(r(t)) = [16t^2, -t^2]$$

$$\vec{F}(r(t)) \cdot \frac{d\vec{r}}{dt} = (16t^2)(1) + (-t^2)(4) = 16t^2 - 4t^2 = 12t^2$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 12t^2 \, dt = 12 \left[\frac{t^3}{3} \right]_0^1 = 4$$

Path Independence / Fundamental Theorem for Line Integrals: If $\vec{F} = \text{grad } g$ (conservative vector field, i.e., $\text{Curl } \vec{F} = \vec{0}$), then:

$$\int_C \vec{F} \cdot d\vec{r} = \int_A^B \text{grad } g \cdot d\vec{r} = g(B) - g(A)$$

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14 Double Integrals and Order of Integration

Double Integral Definition over Region R

$$\iint_R f(x, y) dA = \text{Volume under } z = f(x, y)$$

For rectangular bounds $[a, b] \times [c, d]$:

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx$$

For variable bounds where $y \in [h(x), g(x)]$ and $x \in [a, b]$:

$$\iint_R f(x, y) dA = \int_a^b \int_{h(x)}^{g(x)} f(x, y) dy dx$$

If $f(x, y) = \phi(x)\psi(y)$ on region $[a, b] \times [c, d]$:

$$\iint_R f(x, y) dA = \left[\int_a^b \phi(x) dx \right] \cdot \left[\int_c^d \psi(y) dy \right]$$

Example: Changing Order of Integration

Evaluate $\int_0^2 \int_0^4 (x^2 + y^2) dx dy$:

Direct Integration (dx first):

$$\int_0^2 \left[\frac{x^3}{3} + xy^2 \right]_0^4 dy = \int_0^2 \left(\frac{64}{3} + 4y^2 \right) dy = \left[\frac{64y}{3} + \frac{4y^3}{3} \right]_0^2 = \frac{128}{3} + \frac{32}{3} = \frac{160}{3}$$

Reversed Order (dy first):

$$\begin{aligned} \int_0^4 \int_0^2 (x^2 + y^2) dy dx &= \int_0^4 \left[x^2y + \frac{y^3}{3} \right]_0^2 dx = \int_0^4 \left(2x^2 + \frac{8}{3} \right) dx \\ &= \left[\frac{2x^3}{3} + \frac{8x}{3} \right]_0^4 = \frac{128}{3} + \frac{32}{3} = \frac{160}{3} \end{aligned}$$

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15 Area Computation and Change of Variables

Area of Triangle enclosed by $(0, 0)$, $(b, 0)$, and (b, h)

The hypotenuse is given by line $y = \frac{h}{b}x$.

Using double integral with $x \in [0, b]$ and $y \in [0, \frac{h}{b}x]$:

$$\text{Area } A = \iint_R 1 \, dA = \int_0^b \left(\int_0^{\frac{h}{b}x} dy \right) dx = \int_0^b \frac{h}{b}x \, dx = \frac{h}{b} \left[\frac{x^2}{2} \right]_0^b = \frac{h}{b} \cdot \frac{b^2}{2} = \frac{bh}{2}$$

Change of Coordinates and Jacobian

To transform variables from $(x, y) \rightarrow (r, \theta)$:

$$\iint_R f(x, y) \, dx \, dy = \iint_R f(r, \theta) |J| \, dr \, d\theta$$

where J is the Jacobian determinant:

$$J = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$